

Breast Cancer Classification Using Machine Learning

Overview

This project applies machine learning techniques to classify breast tumors as **benign** or **malignant** using the Wisconsin Breast Cancer Dataset. Three classification models were implemented and compared: Logistic Regression, Random Forest, and Support Vector Machine (SVM).

Dataset

- **Source:** Wisconsin Breast Cancer Dataset
- **Samples:** 569 (357 Benign, 212 Malignant)
- **Features:** 30 numerical features extracted from digitized images of fine needle aspirate (FNA) of breast masses
- **Target:** Diagnosis (Benign = 0, Malignant = 1)

Features Description

Each sample contains three types of measurements for 10 cell nucleus characteristics:

- **_mean:** Mean value
- **_se:** Standard error
- **_worst:** Largest/worst value

Key characteristics include: radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension.

Project Structure

```
breast-cancer-classification/  
├── breast_cancer_classification.ipynb  
├── wisc_bc_data.csv  
└── README.md
```

Methodology

1. Exploratory Data Analysis (EDA)

- Class distribution analysis

- Feature distribution histograms
- Correlation heatmap
- Boxplots comparing benign vs malignant features
- Pairplot of top correlated features

2. Preprocessing

- Removed irrelevant columns (id)
- Encoded target variable
- Train/test split (80/20) with stratification
- Feature scaling using StandardScaler

3. Models

- Logistic Regression (solver: liblinear)
- Random Forest (100 estimators)
- Support Vector Machine (RBF kernel)

Results

Model	Accuracy	Recall (Malignant)
Logistic Regression	97.37%	0.98

Model	Accuracy	Recall (Malignant)
Random Forest	93.86%	0.93
SVM	98.25%	0.98

Key Findings

- **SVM achieved the highest accuracy (98.25%)**
- **Most important features:** `area_worst` and `points_worst`
- From a biological perspective, malignant tumors show larger cell areas and more irregular boundaries – consistent with pathological hallmarks of cancer aggressiveness
- `_worst` features outperformed `_mean` features in predictive power, reflecting that the most abnormal cells drive malignancy

Biological Interpretation

The dominance of `area_worst` and `points_worst` aligns with known cancer biology:

- Malignant cells undergo uncontrolled proliferation, resulting in larger cell sizes
- Irregular concave boundaries reflect loss of normal cell architecture
- These morphological features mirror what pathologists assess under microscopy

Requirements

```
pandas  
numpy  
matplotlib  
seaborn  
scikit-learn
```

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